

The Cohomological Price of NOT

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Abstract

The minimal number of negations needed to compute a Boolean function is governed by Markov’s decrease invariant $d(f)$: circuits need $\lceil \log_2(d(f)+1) \rceil$ negations (Markov 1958), and formulas need exactly $d(f)$ (Morizumi 2009). We show that the decrease invariant carries a *cohomological address*. On the dual-rail four-valued lattice, the negation σ and the rail-swap P generate a Klein four-group V_4 whose character group $\text{Hom}(V_4, \mathbb{F}_2) \cong \mathbb{F}_2^2$ has exactly three nonzero characters, and χ_σ is the *unique* one that prices negation — the two alternatives fail in opposite directions, witnessed by P (face-flip 1, cost 0) and σP (face-flip 0, cost 1). The main theorem reads the full invariant as an iterated extension: $d(f)$ equals the minimal length of a nested χ_σ -tower, each rung adjoining exactly one negation inside a monotone context. The upper bound uses a single uniform three-zone witness; the lower bound is an alternation argument valid over any finite poset. All statements are machine-verified in core Lean 4 with kernel axioms `propext` and `Quot.sound` only.

1 Introduction

Markov [1] determined the inversion complexity of Boolean circuits: $I(f) = \lceil \log_2(d(f) + 1) \rceil$, where the decrease $d(f)$ counts the maximal number of $1 \rightarrow 0$ descents of f along an increasing chain of inputs. Morizumi [2] settled the formula model: $I_{\text{for}}(f) = d(f)$ exactly. For multi-valued logic, Kochergin–Mikhailovich [7] extended the *circuit* law (still logarithmic); no formula-exact multi-valued law appears in the literature.

This note contributes three things.

(i) **A four-valued frame.** On the dual-rail lattice **D4** the exact formula-model law persists (the companion preprint [10] proves $\nu(f) = d(f)$ in that setting; we do not reprove it here).

(ii) **A cohomological reading.** The negation σ and the monotone rail-swap P generate V_4 , and among the three nonzero characters of $H^1(V_4; \mathbb{F}_2)$ exactly one — χ_σ — prices negation (§3). The seemingly natural candidate, the face-flip character $\chi_\sigma + \chi_P$, fails in *both* directions.

(iii) **A nested normal form.** $d(f)$ is the minimal length of a nested tower $T_0 \subseteq T_1 \subseteq \dots$ in which each rung adjoins exactly one negation inside a monotone context (§6). Morizumi’s upper-bound construction is the *parallel* decomposition $f_1 \vee (f_2 \wedge \neg f_t)$; the nested single-peel form proved here is strictly more structured, and appears to be new even in the Boolean case. The upper bound is produced by one uniform three-zone witness (§5), whose correctness is a short budget argument; a cautionary example shows the obvious witness is wrong already at $d(f) = 1$.

Everything below is machine-verified in core Lean 4 (§7).

2 Preliminaries

Let $R = \{0, 1\}$ with $0 < 1$, and order R^n componentwise. A *chain* is an increasing sequence $x^0 < x^1 < \dots < x^m$ in R^n . For $f : R^n \rightarrow R$ and a chain X , the *decrease* $d_X(f)$ is the number of indices i with $f(x^i) = 1$ and $f(x^{i+1}) = 0$; $d(f) = \max_X d_X(f)$ over maximal chains. For a binary word w , write $\text{rises}(w)$ and $\text{drops}(w)$ for the numbers of 01- and 10-adjacencies. The *decrease potential* is

$$\omega(x) := \max\{\text{drops of } f \text{ along an increasing chain ending at } x\},$$

so ω is nondecreasing along chains, $\omega \leq d(f)$ everywhere, and every descent step strictly increases ω .

The dual-rail lattice is $\mathbf{D4} = \{0, 1\}^2$, ordered componentwise (four values: bottom $(0, 0)$, the two *resolved* values $(1, 0)$ and $(0, 1)$ forming the face R , and top $(1, 1)$). Define the involutions $\sigma(a, b) = (1 - a, 1 - b)$ (negation; order-reversing) and $P(a, b) = (b, a)$ (rail-swap; monotone). They commute and generate $V_4 = \{\text{id}, \sigma, P, \sigma P\}$.

3 The pricing character

$H^1(V_4; \mathbb{F}_2) = \text{Hom}(V_4, \mathbb{F}_2) \cong \mathbb{F}_2^2$, with the three nonzero characters determined by their kernels: χ_σ (kernel $\{\text{id}, P\}$), χ_P (kernel $\{\text{id}, \sigma\}$), and the face-flip character $\chi_\sigma + \chi_P$ (kernel $\{\text{id}, \sigma P\}$) — the character that is 1 exactly on the elements moving the resolved face R . For $\gamma \in V_4$ let $\nu(\gamma) \in \{0, 1\}$ be its negation cost: 0 if γ is monotone, 1 otherwise.

Proposition 3.1 (gate typing). *As vectors over $(\text{id}, \sigma, P, \sigma P)$:*

γ	<i>monotone</i>	$\nu(\gamma)$	$\chi_\sigma(\gamma)$	$(\chi_\sigma + \chi_P)(\gamma)$
id	<i>yes</i>	0	0	0
σ	<i>no</i>	1	1	1
P	<i>yes</i>	0	0	1
σP	<i>no</i>	1	1	0

Hence $\nu = \chi_\sigma$ pointwise on V_4 , and χ_σ is the **unique** nonzero character with this property: χ_P fails at σ , and the face-flip character fails in both directions — at P (flip without cost) and at σP (cost without flip).

Proof. Inspection of the four actions on $\mathbf{D4}$: σ swaps both the endpoints $(0, 0) \leftrightarrow (1, 1)$ and the face $(1, 0) \leftrightarrow (0, 1)$ and reverses the order; P fixes the endpoints, swaps the face, and is monotone; σP swaps the endpoints, fixes the face, and is not monotone (it exchanges bottom and top). Machine-checked as `gate_table` and `chiSigma_eq_not_kMono` (axiom-free kernel `decide`). $\square$

The two witnesses deserve names: P is the *first separating witness* (face-flip 1, cost 0) and σP the *second* (cost 1, face-flip 0). Together they show that “pricing negation” and “moving the resolved face” are different characters — they differ exactly by χ_P .

4 Subadditivity and the lower bound

Lemma 4.1 (alternation). *For every binary word w , $\text{rises}(w) \leq \text{drops}(w) + 1$; if w begins with 1, then $\text{rises}(w) \leq \text{drops}(w)$.*

Proof. Induction on the word, tracking the first letter: each maximal 1-block contributes at most one rise on its left and one drop on its right, and only the initial 1-block can lack a preceding rise. (Lean: `rises_le_drops_succ`.) $\square$

Lemma 4.2 (drop transfer). *Let g be monotone in both arguments, h arbitrary, and $f(x) = g(x, \neg h(x))$. If $x < x'$ is a chain step with $f(x) = 1$ and $f(x') = 0$, then $h(x) = 0$ and $h(x') = 1$ (an ascent of h).*

Proof. If $\neg h(x) \leq \neg h(x')$ then $(x, \neg h(x)) \leq (x', \neg h(x'))$ and monotonicity of g forces $f(x) \leq f(x')$, a contradiction. So $\neg h$ descends, i.e. h ascends. $\square$

Theorem 4.3 (subadditivity). *For monotone g and arbitrary h , $d(g(x, \neg h(x))) \leq d(h) + 1$. In fact, along any single chain, $\text{drops}(f\text{-word}) \leq \text{rises}(h\text{-word}) \leq \text{drops}(h\text{-word}) + 1$; the argument uses only adjacent steps, so it is valid over any relation, in particular any finite poset. (Lean: `subadd_chain`, `subadd_decOn`.)*

Proof. Each descent of f is matched by an ascent of h at the same step (Lemma 4.2); ascents are bounded by descents plus one (Lemma 4.1); descents of h along any chain are at most $d(h)$. $\square$

Corollary 4.4 (tower lower bound). *Define $T_0 = \text{the monotone functions}$ and $T_{k+1} = \{g(x, \neg h(x)) : g \text{ monotone, } h \in T_k\}$. Then $f \in T_k$ implies $d(f) \leq k$. (Lean: `F1_lower`, via the end-value-refined form $\text{rises}(w) \leq \text{drops}(w) + [w \text{ ends in } 1]$.)*

Corollary 4.5 (no skipping). *If $f = g(x, \neg h)$ with g monotone, then $d(h) \geq d(f) - 1$.*

5 The three-zone witness

Fix f with $d(f) = k \geq 1$. Call a pair $x < x'$ *violated* if $f(x) = 1$ and $f(x') = 0$; write V^- and V^+ for the lower and upper points of violated pairs.

Lemma 5.1 (potential jump). *Across a violated pair, $\omega(x) + 1 \leq \omega(x')$. In particular $\omega(x) \leq k - 1$ and $\omega(x') \geq 1$. (Lean: `omega_viol`.)*

Proof. Take a cover-path from x up to x' and induct along it. If the first step already descends ($f = 1$ at its bottom, $f = 0$ at its top), strictness of ω across a descent gives the jump, and monotonicity of ω carries it to x' . Otherwise f is still 1 at the intermediate point and the induction hypothesis applies from there, with monotonicity of ω across the first step. $\square$

Theorem 5.2 (three-zone witness). *Define*

$$h := 1 \text{ on } \{\omega \geq k\}, \quad \neg f \text{ on } \{1 \leq \omega \leq k - 1\}, \quad 0 \text{ on } \{\omega = 0\}.$$

Then (a) h covers every violated pair: $h = 0$ on V^- and $h = 1$ on V^+ ; and (b) $d(h) = k - 1$ exactly. (Lean: `hgen_covering`, `hgen_drops_le`, `omega_hgen_le`, over an abstract descent potential; `cube_descPotential` discharges the potential axioms for the concrete ω .)

Proof. (a) For $x \in V^-$: $\omega(x) \leq k - 1$ by Lemma 5.1, so $h(x)$ is either 0 (zone 0) or $\neg f(x) = 0$ (middle zone). For $x' \in V^+$: $\omega(x') \geq 1$, so $h(x')$ is either $\neg f(x') = 1$ (middle) or 1 (top zone).

(b) Along any chain, ω is nondecreasing, so the chain traverses the zones in the order $\{\omega = 0\} \rightarrow \{1 \leq \omega \leq k - 1\} \rightarrow \{\omega \geq k\}$, each a contiguous (possibly empty) stretch. h is constant 0, then $\neg f$, then constant 1. A descent of h cannot start at the 0-zone ($h = 0$), cannot occur inside the top zone ($h = 1$), and cannot occur at either zone boundary (entering the middle from 0 cannot descend; leaving the middle to the top ends at 1). So every descent of h lies inside the middle stretch and is an *ascent* of f there. Inside the middle stretch, each descent of f strictly increases ω , and ω stays within $[1, k - 1]$; hence the stretch contains at most $k - 2$ descents of f , and by alternation at most $k - 1$ ascents. So $d(h) \leq k - 1$; equality follows from Corollary 4.5 applied to the monotone-context factorization produced in §6. $\square$

Remark 5.3 (the naive witness fails). The tempting witness $h_1 := \neg f \wedge [\omega \geq 1]$ is wrong already at $k = 1$. On three variables take $f(010) = f(111) = 1$ and $f = 0$ elsewhere: then $d(f) = 1$, but along the chain $000 < 100 < 110 < 111$ the potential jumps to 1 at 110 through the *side* chain $000 < 010 < 110$, so h_1 takes values 0, 0, 1, 0 — one descent, though $k - 1 = 0$. The top-zone cap ($h \equiv 1$ on $\{\omega \geq k\}$) is exactly what repairs this. The example generalizes: side-chains can raise ω without any descent on the chain being traversed.

6 The nested tower theorem

Lemma 6.1 (monotone extension). *Let h cover every violated pair of f . Then there is a monotone g with $f(x) = g(x, \neg h(x))$: define*

$$g(y, v) := 1 \iff \exists x \leq y \ (h(x) = 1 \text{ or } v = 1) \wedge f(x) = 1,$$

a monotone function of (y, v) ; the covering hypothesis rules out the only conflict (a witness $x \leq y$ with $f(x) = 1$ forcing $g = 1$ while $f(y) = 0$ — such a pair is violated, so $h(x) = 0$ and $h(y) = 1$, and the disjunct $h(x) \vee \neg h(y)$ fails). No choice principle is needed — the extension is an explicit finite disjunction. (Lean: `gExt`, `gExt_mono`, `gExt_agrees`.)

Theorem 6.2 (nested tower). *For every nonconstant $f : R^n \rightarrow R$,*

$$f \in T_k \iff d(f) \leq k.$$

Hence $d(f)$ is the minimal length of a nested χ_σ -tower over the monotone clone, each rung consuming exactly one χ_σ -twist. (Lean: `tower_of_decPts` for the upper bound — constructive, producing the witness of §5 at each rung — and `F1_lower` for the lower bound.)

Proof. ($\Leftarrow$, upper bound) Induction on k . If $d(f) = 0$, f is monotone. Otherwise let h be the three-zone witness for $K := d(f)$: by Theorem 5.2(b) $d(h) = K - 1$, by induction $h \in T_{K-1}$, and by Lemma 6.1 with Theorem 5.2(a) $f = g(x, \neg h)$ for a monotone g , so $f \in T_K \subseteq T_k$. ($\Rightarrow$ is Corollary 4.4.) $\square$

Remark 6.3 (models). In the formula model this recovers $I_{\text{for}}(f) = d(f)$ [2] with a *nested* optimal form: Morizumi’s construction is the parallel decomposition $f_1 \vee (f_2 \wedge \neg f_t)$, in which the recursive part and the fresh negation sit in separate branches; Theorem 6.2 shows a single nested branch suffices. In the circuit model the tower reading diverges from $I(f)$ once $d \geq 3$, by Markov’s logarithmic law — the tower is a formula-model object.

Remark 6.4 (level, not degree). The tower index is a filtration level at fixed cohomological degree one. No identification with any higher-degree or higher-level operator theory is claimed.

7 Mechanization

All statements are verified in core Lean 4 (no mathlib), 19 theorems; the axiom print for every theorem is `[propext, Quot.sound]`, and the finite character tables (`gate_table`, `chiSigma.eq_not_kMono`) depend on **no axioms**. There is no `native_decide` and no `sorry`; the upper bound is constructive (no choice axiom). Correspondence:

Paper	Lean (lean/DecBridge/)
Lemma 4.1	Words: <code>rises_le_drops_succ</code>
Lemma 4.2	Subadd: <code>drop_transfer</code>
Theorem 4.3	Subadd: <code>subadd_chain</code> , <code>subadd_decOn</code>
Proposition 3.1	Gates: <code>gate_table</code> , <code>chiSigma.eq_not_kMono</code>
Theorem 5.2 (abstract)	Hgen: <code>hgen_covering</code> , <code>hgen_drops_le</code>
ω is a descent potential	Cube: <code>cube_descPotential</code>
Lemma 5.1	CubeViol: <code>omega_viol</code>
$\omega = \text{chain-max}$ (both directions)	CubeDec: <code>chainDrops_le_omega</code> , <code>omega_attained</code>
Theorem 5.2(b) exact	CubeDec: <code>omega_hgen_le</code>
Lemma 6.1	Tower: <code>gExt</code> , <code>gExt_mono</code> , <code>gExt_agrees</code>
Theorem 6.2 upper	Tower: <code>tower_of_decPts</code>
Cor. 4.4 / Thm. 6.2 lower	Tower: <code>F1_lower</code> , <code>decPts_le_of_tower</code>

The Lean development is slightly *more* general than the text: subadditivity is proved for chains of an arbitrary relation (no order axioms), and the witness theorem over an abstract descent potential, of which the cube ω (defined by weight-fuel recursion over lower covers) is an instance. The upper bound is stated over an explicit finite point family (agreement on the listed points), which is the meaningful content on the cube. Independent stdlib-only Python verifications (exhaustive $n \leq 4$; randomized $n = 5$) accompany the artifact.

Authorship and provenance

Produced by *KoreoLoop*, an autonomous multi-agent research loop developed and operated by the author, running frontier language models (Anthropic Claude family) for discovery, formalization, and adversarial verification, with the **Lean 4 kernel as the final acceptance gate**. The author directed the research programme and verified the pipeline. In line with the author’s research-ethics policy, no claim of academic priority is made beyond full disclosure of how the work was produced. Corrections are invited. Artifact: <https://github.com/ycmath/cohomological-price-of-not>.

References

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